

Topic B8

Transport in plants

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B8.1	Xylem and phloem
B8.2	Water uptake
B8.3	Transpiration
B8.4	Translocation

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Xylem

Xylem vessels are dead, hollow tubes with lignified walls. They carry water and mineral ions upward and also support the plant. No end walls and a continuous lumen reduce resistance; lignin prevents collapse under tension.

Examiner insight

Avoid describing xylem transport as active pumping by the vessel.

2. Water uptake

Root hair cells provide a large surface area. Water enters by osmosis from higher soil water potential to lower cell water potential. Mineral ions may enter by active transport. Water crosses the cortex and enters xylem, maintaining a continuous column.

Examiner insight

Separate the mechanisms for water and mineral-ion uptake.

3. Transpiration

Transpiration is loss of water vapour from leaves. Water evaporates from mesophyll cell surfaces, diffuses through air spaces and exits stomata. This creates tension that pulls water through xylem. Rate rises with light, temperature and wind, and falls with high humidity.

Examiner insight

Explain each environmental effect through evaporation or the water-vapour gradient.

4. Translocation

Phloem transports sucrose and amino acids between sources and sinks, in either direction in different tubes. Sources release assimilates, such as photosynthesising leaves; sinks use or store them, such as roots, fruits and growing tissues. Translocation requires living tissue and energy.

Examiner insight

Phloem is not simply “downward transport”; direction depends on source and sink.

Worked application

Situation: Stomata close during a hot dry afternoon.

Prediction: Water loss decreases but photosynthesis may fall.

Reason: Smaller stomatal opening reduces diffusion of water vapour and also limits carbon dioxide entry.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B8

ransport in plants

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines transpiration? [1]
 - A loss of water vapour from plant leaves
 - B movement of sucrose and amino acids in phloem
 - C tissue transporting water and mineral ions and providing support
 - D living tissue transporting assimilates between sources and sinks
2. Which statement best defines translocation? [1]
 - A tissue transporting water and mineral ions and providing support
 - B movement of sucrose and amino acids in phloem
 - C living tissue transporting assimilates between sources and sinks
 - D a region releasing sucrose into phloem
3. Which statement best defines xylem? [1]
 - A living tissue transporting assimilates between sources and sinks
 - B a region releasing sucrose into phloem
 - C tissue transporting water and mineral ions and providing support
 - D loss of water vapour from plant leaves
4. Which statement best defines phloem? [1]
 - A a region releasing sucrose into phloem
 - B loss of water vapour from plant leaves
 - C movement of sucrose and amino acids in phloem
 - D living tissue transporting assimilates between sources and sinks
5. Which statement best defines source? [1]
 - A a region releasing sucrose into phloem
 - B loss of water vapour from plant leaves
 - C movement of sucrose and amino acids in phloem
 - D tissue transporting water and mineral ions and providing support
6. Which statement about xylem is correct? [1]
 - A Movement of sucrose through phloem.
 - B Xylem vessels are dead, hollow tubes with lignified walls.
 - C Movement of water in xylem.
 - D Living tissue transporting sugars.
7. Which statement about water uptake is correct? [1]
 - A Movement of water in xylem.
 - B Living tissue transporting sugars.
 - C Root hair cells provide a large surface area.
 - D Dead hollow vessels carrying water only upward.

8. Which statement about transpiration is correct? [1]
- A Living tissue transporting sugars.
 - B Dead hollow vessels carrying water only upward.
 - C Any organ below a leaf.
 - D Transpiration is loss of water vapour from leaves.
9. Which statement about translocation is correct? [1]
- A Phloem transports sucrose and amino acids between sources and sinks, in either direction in different tubes.
 - B Dead hollow vessels carrying water only upward.
 - C Any organ below a leaf.
 - D Movement of sucrose through phloem.
10. Which prediction is most appropriate when stomata close during a hot dry afternoon? [1]
- A Movement of sucrose through phloem.
 - B Living tissue transporting sugars.
 - C Water loss decreases but photosynthesis may fall.
 - D Any organ below a leaf.

Section B · Structured questions

11 Biologists use precise vocabulary when studying transport in plants.

(a) Define transpiration. [2]

(b) Define translocation. [2]

(c) Distinguish xylem from the incorrect idea: “living tissue transporting sugars”. [2]

(d) State two important features of xylem. [2]

[Total: 8]

12 Stomata close during a hot dry afternoon.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records distance per minute under three conditions.

condition	distance per minute
still air	2
gentle wind	7
strong wind	12

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to use a potometer to test wind speed.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of transport in plants to an unfamiliar context.

(a) Define phloem. [2]

(b) Explain the biological importance of transpiration. [3]

(c) Correct this statement: “any organ below a leaf”.

[2]

(d) State one further question that could be investigated.

[1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Loss of water vapour from plant leaves.
- 2 B. Movement of sucrose and amino acids in phloem.
- 3 C. Tissue transporting water and mineral ions and providing support.
- 4 D. Living tissue transporting assimilates between sources and sinks.
- 5 A. A region releasing sucrose into phloem.
- 6 B. Avoid describing xylem transport as active pumping by the vessel.
- 7 C. Separate the mechanisms for water and mineral-ion uptake.
- 8 D. Explain each environmental effect through evaporation or the water-vapour gradient.
- 9 A. Phloem is not simply “downward transport”; direction depends on source and sink.
- 10 C. Smaller stomatal opening reduces diffusion of water vapour and also limits carbon dioxide entry.

Solutions · Structured questions

11. (a) loss of water vapour from plant leaves [2]. (b) movement of sucrose and amino acids in phloem [2]. (c) tissue transporting water and mineral ions and providing support; not living tissue transporting sugars [2]. (d) Credit any two accurate features from the guide: Xylem vessels are dead, hollow tubes with lignified walls. They carry water and mineral ions upward and also support the plant. No end walls and a continuous lumen reduce resistance; lignin prevents collapse under tension. [2].

12. (a) Water loss decreases but photosynthesis may fall [2]. (b) Smaller stomatal opening reduces diffusion of water vapour and also limits carbon dioxide entry [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $12 - 2 = 10$ [2]. (b) The values change from 2 to 12, with the highest value 12. [2]. (c) Wind removes humid air and maintains a steep water-vapour gradient [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Wind speed [1]. (b) Bubble distance per time [1]. (c) Same shoot, temperature, light and humidity [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Seal joints, cut shoot under water and repeat [1].

15. (a) living tissue transporting assimilates between sources and sinks [2]. (b) Award linked points from: Transpiration is loss of water vapour from leaves. Water evaporates from mesophyll cell surfaces, diffuses through air spaces and exits stomata. This creates tension that pulls water through xylem. Rate rises with light, temperature and wind, and falls with high humidity. [3]. (c) Correct statement: source means a region releasing sucrose into phloem [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.